

CropPlanner: An AI-Powered Web Application for Crop Planning Recommendations

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Abstract—Agricultural price volatility exceeding 20% annually, combined with the prevalence of monoculture farming practices, exposes Thai farmers to significant income risks. This paper presents CropPlanner, an integrated system comprising three components: (1) a comparative study of AI models for 12-month crop price forecasting, including LSTM, Transformer, ARIMA, and a proposed Rolling-Window LSTM; (2) a crop planning recommendation system formulated as a Weighted Job Scheduling Problem solved by a Greedy Heuristic; and (3) a web application for farmers. Experiments were conducted using 20 years of monthly price data (2004–2024) for four major field crops—cassava, corn, green bean, and soybean—in Northeastern Thailand. Results show that no single model dominates across all crops: Standard LSTM achieved the highest average accuracy of 92.93%, while Rolling-Window LSTM—with window sizes selected via a held-out validation year to prevent data leakage—achieved correct peak timing for two of four crops and within one month for a third, a property critical for scheduling. Exhaustive enumeration over all $4! = 24$ permutations identified the optimal non-overlapping four-crop rotation plan yielding 39,212 THB/rai/year, approximately 1.9 times the income of the best monoculture option, while revealing that the greedy heuristic is suboptimal by 14.8%.

Index Terms—Machine Learning, Price Forecasting, Web Application, Time Series, ARIMA, LSTM, Transformer, Crop Planning

This paper presents CropPlanner, an end-to-end system linking AI-based price forecasting with crop scheduling and a farmer-facing web application. Our contributions are threefold:

- 1) **Model comparison and improvement:** We compare LSTM, Transformer, ARIMA, and a proposed Rolling-Window LSTM for 12-month price forecasting of four Thai field crops, analyzing both overall accuracy and peak timing accuracy.
- 2) **Crop planning recommendation:** We formulate planting schedule optimization as a Weighted Job Scheduling Problem and solve it with a Greedy Heuristic that produces non-overlapping planting–harvest timelines.
- 3) **Web application:** We develop CropPlanner, a prototype web application that presents forecasting results and planting recommendations through intuitive visualizations.



Fig. 1. System overview of CropPlanner comprising three integrated components.

I. INTRODUCTION

Agriculture forms a critical foundation of the Thai economy. Four major field crops—cassava, corn, green bean, and soybean—account for over 60% of cultivated land and generate export revenues exceeding US\$13 billion annually [1]–[4]. However, prices of these crops exhibit volatility exceeding 20% within a 12-month period [5], creating substantial income risks for farmers.

A key contributing factor is the widespread practice of monoculture farming. Many farmers plant the same crop continuously throughout the year, leaving them vulnerable when prices decline during certain periods [6]. While government price support programs provide temporary relief, they are insufficient for long-term income stabilization. What farmers fundamentally lack are tools for forward-looking price forecasting and systematic planting schedule optimization.

Recent advances in machine learning, particularly time-series models such as LSTM and Transformer, have demonstrated the ability to reduce Mean Absolute Error (MAE) in agricultural price prediction by up to 35% compared to traditional methods [7]. Yet most existing work focuses on forecasting individual crop prices in isolation, without connecting predictions to actionable planting recommendations [8].

II. RELATED WORK

A. Time-Series Forecasting for Agricultural Prices

Agricultural price forecasting has been extensively studied using statistical and machine learning approaches. Traditional ARIMA models remain widely used for linear time-series data but struggle with structural breaks and nonlinear patterns. Deep learning models have gained prominence, with LSTM networks—designed to capture long-term dependencies through gating mechanisms [9]—and Transformer architectures—leveraging self-attention to process entire sequences simultaneously [10]—showing superior performance on various forecasting benchmarks.

Rohan [7] demonstrated that LSTM and Transformer models can reduce MAE by up to 35% compared to traditional methods for crop price prediction. Javaid et al. [8] surveyed AI applications in agriculture broadly, including predictive analytics. However, most prior work forecasts a single crop in isolation and does not connect predictions to downstream planning systems.

B. Crop Scheduling and Decision Support Systems

Resource allocation in agriculture can be modeled as optimization problems such as the Knapsack Problem for area allocation or Job Scheduling for temporal planning. Sarker and Ray [12] formulated crop planning as a multi-objective optimization model and demonstrated the use of evolutionary algorithms to identify Pareto-optimal planting configurations that balance net return against crop diversity. Their work highlights the applicability of combinatorial optimization to agricultural scheduling, though it does not incorporate price forecasting as a decision input.

Agricultural Decision Support Systems (DSS) have been developed in various contexts. Zhai et al. [13] surveyed thirteen representative DSS for Agriculture 4.0, finding that most systems focus on individual sub-tasks—irrigation management, pest control, or nutrient planning—and typically rely on static agronomic data such as planting calendars or soil information rather than AI-driven price forecasts as decision variables.

The gap that this work addresses is the integration of AI price forecasting with scheduling optimization and farmer-accessible web delivery—a pipeline that has received limited attention in prior literature.

III. DATA AND PREPROCESSING

A. Data Sources

Price data were obtained from two Thai government agencies: the Department of Internal Trade (DIT) and the Office of Agricultural Economics (OAE). Table I summarizes the dataset.

TABLE I
DATASET SUMMARY

Property	Value
Crops	Cassava, Corn, Green Bean, Soybean
Region	Northeastern Thailand
Period	Jan 2004 – Dec 2024 (20 years)
Frequency	Monthly
Records	~252 per crop
Currency	Thai Baht (THB/kg)

B. Preprocessing

Raw data in PDF and Excel format with Thai Buddhist Era dates underwent several cleaning steps: date conversion from Buddhist Era (BE) to Common Era (CE), duplicate removal, and unit normalization. Processing was performed using Python with Pandas and NumPy. The final cleaned data was stored in long-format CSV files with date and price columns.

C. Train–Validation–Test Split

Data were divided into three partitions: a training set (2004–2022), a validation set (2023), and a test set (2024). The scaler was fit exclusively on training data to prevent data leakage. The validation set was used solely for hyperparameter selection—specifically, the Rolling-Window LSTM window size (Section IV-A4)—ensuring that the test set remained entirely unseen during model development. For the final

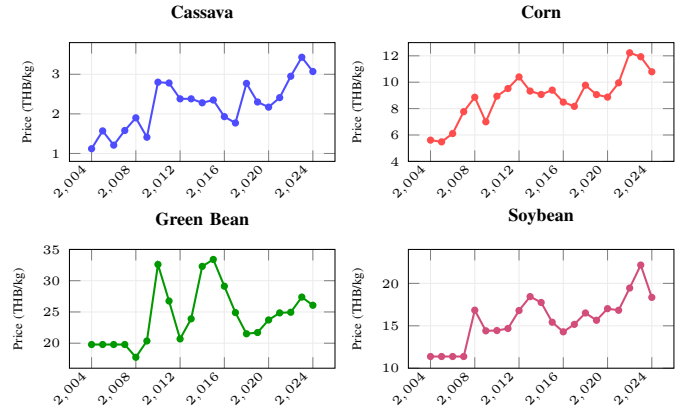


Fig. 2. Yearly average prices (2004–2024) for four Thai field crops showing long-term trends, structural breaks, and the distinct price scales of each commodity.

evaluation, models were retrained on the combined training and validation data (2004–2023) and evaluated on the held-out test year (2024).

IV. METHODOLOGY

The CropPlanner system comprises three components: (A) AI price forecasting models, (B) crop planning recommendation, and (C) a web application. All models take 12 months of historical prices as input and produce 12-month forecasts.

A. AI Price Forecasting Models

Table II summarizes the hyperparameters of all four models.

TABLE II
HYPERPARAMETER SUMMARY

Parameter	LSTM	Trans.	ARIMA	RW-LSTM
Architecture	2-layer	2-layer enc.	Auto	3-layer
Hidden / d_{model}	32	64	auto	32
Attn. heads	—	8	—	—
FF dim	—	128	—	—
Epochs	300	500	—	300
Learning rate	0.01	0.001	—	0.01
Loss function	MSE	MSE	—	MSE
Library	PyTorch	PyTorch	StatsFC	PyTorch

1) *Standard LSTM*: Long Short-Term Memory (LSTM) [9] is a recurrent neural network designed to learn long-term dependencies via gating mechanisms (forget, input, and output gates). Our model consists of 2 layers with hidden size 32, receiving 12 months of input and producing 12 months of output. Training used MSE loss on PyTorch for 300 epochs with learning rate 0.01 and random seed 42.

2) *Transformer*: The Transformer model [10] uses multi-head self-attention to process the entire sequence simultaneously, efficiently capturing global dependencies across time steps. Our encoder-only configuration uses $d_{\text{model}} = 64$, 8 attention heads, 2 layers, and feed-forward dimension 128 with positional encoding. Training ran for 500 epochs at a learning rate of 0.001.

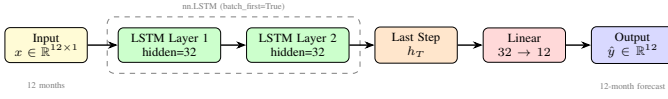


Fig. 3. Standard LSTM architecture. Input sequence of 12 monthly prices passes through 2 LSTM layers (hidden=32). The last time step's hidden state is projected to 12 output values via a linear layer.

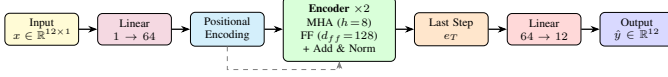


Fig. 4. Transformer architecture (encoder-only). Prices are projected to $d_{\text{model}} = 64$, augmented with positional encoding, and processed by 2 encoder layers with 8-head attention. The last position's encoding is decoded to 12 output values.

3) *ARIMA*: ARIMA (Autoregressive Integrated Moving Average) [11] combines autoregressive, differencing, and moving average components. We used AutoARIMA from the StatsForecast library, which automatically selects (p, d, q) parameters through time-series decomposition into trend, seasonality, and residual components.

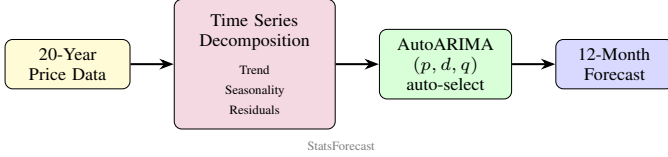


Fig. 5. ARIMA forecasting pipeline. Historical data is decomposed into trend, seasonality, and residual components before AutoARIMA selects optimal (p, d, q) parameters.

4) *Rolling-Window LSTM (Proposed Improvement)*: Analysis of the standard LSTM revealed three issues: (1) off-peak prediction—the model misidentified peak price months; (2) limited long-term dependency over the full 20-year sequence; and (3) sensitivity to noisy data.

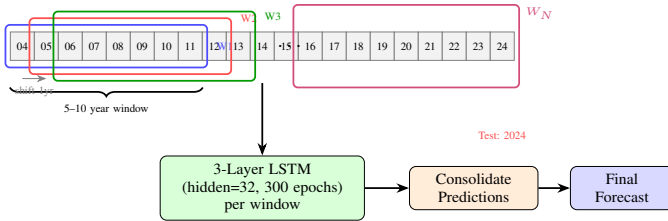


Fig. 6. Rolling-Window LSTM training concept. The dataset is divided into overlapping windows of 5–10 years (selected per crop via validation), each shifted by 1 year. A 3-layer LSTM is trained on each window independently; predictions are consolidated into a final forecast.

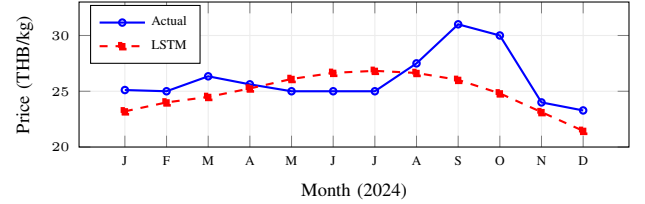
To address these issues, we propose the Rolling-Window LSTM. The dataset is divided into rolling windows of 5–10 years (crop-dependent, selected via validation), each shifted by 1 year. A 3-layer LSTM (hidden size 32, 300 epochs) is trained on each window independently. Predictions from all windows are then consolidated into a final forecast for the target year.

Window sizes were selected via validation-set evaluation. For each crop, we swept window sizes from 5 to 10 years and selected the size yielding the lowest MAPE on the 2023 validation year. Table III presents the validation results. Cassava achieved the best validation MAPE with an 8-year window (31.25%), corn with a 6-year window (11.43%), while green bean and soybean performed best with 5-year windows (12.67% and 15.62%, respectively). The final models were then retrained on the combined training and validation data (2004–2023) using these selected window sizes and evaluated on the held-out test year (2024). By learning from multiple shorter windows, the model captures recurring seasonal patterns more accurately—particularly the location of peak prices, which is critical for the downstream scheduling system.

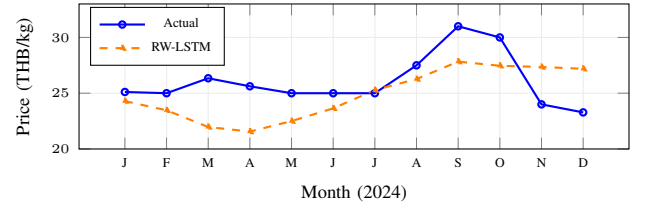
TABLE III
VALIDATION MAPE (%) BY WINDOW SIZE (YEAR 2023)

Crop	5yr	6yr	7yr	8yr	9yr	10yr
Cassava	35.09	31.31	33.15	31.25	31.53	32.03
Corn	13.08	11.43	14.23	11.44	13.60	20.27
Green Bean	12.67	13.34	16.20	14.39	14.96	21.47
Soybean	15.62	20.48	19.15	23.72	25.07	21.73

Bold values indicate the selected window size per crop (lowest validation MAPE).



(a) Standard LSTM (peak: Jul, actual: Sep, $\Delta=2$ mo.)



(b) RW-LSTM (peak: Sep, actual: Sep, $\Delta=0$ mo.)

Fig. 7. Green bean 2024 forecast comparison. (a) Standard LSTM produces a smooth curve peaking in Jul, missing the sharp Sep peak. (b) RW-LSTM correctly predicts peak timing in Sep ($\Delta=0$ mo.) with the validated 5-year window.

5) *Evaluation Metrics*: Model performance was evaluated using three metrics.

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |\hat{y}_t - y_t| \quad (1)$$

Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{\hat{y}_t - y_t}{y_t} \right| \quad (2)$$

Forecast Accuracy¹:

$$\text{Forecast Accuracy} = 1 - \text{MAPE} \quad (3)$$

Additionally, **Peak Timing Accuracy** was measured for the Rolling-Window LSTM, defined as the absolute difference in months between the predicted and actual peak price positions. This metric is critical because the scheduling system uses peak positions to determine harvest dates.

B. Crop Planning Recommendation System

The crop planning problem is formulated as a *Weighted Job Scheduling Problem*. Each crop constitutes a job with a duration equal to its growing period and a weight equal to its expected income. The constraint is that no two jobs may overlap on a single plot of land (1 rai = 1,600 m²).

1) *Phase 1: Job Definition from Forecasts*: Rolling-Window LSTM forecasts are normalized by dividing each crop's price series by its mean, enabling cross-crop comparison. The peak normalized price date (argmax) defines the harvest date—the optimal selling time. The planting date is then computed by subtracting the growing period:

$$t_{\text{plant}} = t_{\text{harvest}} - d_{\text{grow}} \quad (4)$$

Table IV lists the agronomic constants used.

TABLE IV
AGRONOMIC CONSTANTS PER CROP

Crop	Growing Days	Yield (kg/rai)
Cassava	250	3,500
Corn	120	1,500
Green Bean	80	245
Soybean	90	300

Each crop's expected income is computed as:

$$I_i = Y_i \times P_i^{\text{peak}} \times A \quad (5)$$

where Y_i is yield (kg/rai), P_i^{peak} is the price at peak (THB/kg), and A is area. In this study, $A = 1$ rai for all computations. Yield values are national agronomic averages from OAE data and are treated as deterministic constants. This simplification is discussed in Section V-B2.

2) *Phase 2: Greedy Scheduling*: Since independently optimal positions for each crop may overlap, a scheduling algorithm is required. Fig. 8 presents the greedy scheduling process as a flowchart.

The algorithm enforces four constraints: (1) non-overlap—one crop per plot at any time; (2) one planting per crop; (3) harvest must fall within the target year; (4) fixed area of 1 rai. As a greedy heuristic, it does not guarantee global optimality—scheduling a high-income crop first may block two medium-income crops with higher combined revenue. To verify solution quality, we additionally perform an exhaustive search over all $4! = 24$ permutations. With only four crops, exhaustive

enumeration completes in negligible time, serving as an exact optimality benchmark. The greedy heuristic is presented as the general-purpose algorithm applicable to larger crop sets where exhaustive search becomes infeasible.

C. Web Application

The CropPlanner web application was built with ReactJS and Bootstrap 5 for the frontend, Next.js with MySQL for the backend, prototyped in Figma, and deployed on Vercel. It provides two main features: (1) **Crop Prices**—displaying AI-predicted price charts for all four crops; and (2) **Crop Planning**—accepting start month and area inputs, then presenting a Gantt chart recommendation with financial summaries (income, expenses, profit, and ROI).

V. RESULTS

A. Model Comparison

Tables V and VI present MAPE and Accuracy results for the 2024 test year. Bold values indicate the best result per crop.

TABLE V
MAPE (%) FOR 2024 TEST YEAR

Crop	LSTM	Trans.	ARIMA	RW-LSTM
Cassava	4.81	9.32	24.71	23.18
Corn	6.43	15.09	8.10	13.32
Green Bean	7.20	6.34	6.08	9.36
Soybean	9.44	5.71	6.99	22.38
Average	7.07	9.12	11.47	17.06

TABLE VI
FORECAST ACCURACY (%) FOR 2024 TEST YEAR

Crop	LSTM	Trans.	ARIMA	RW-LSTM
Cassava	95.33	89.99	77.74	75.80
Corn	93.29	84.29	92.03	87.02
Green Bean	92.51	93.50	93.45	90.69
Soybean	90.57	94.18	92.82	77.90
Average	92.93	90.49	89.01	82.85

No single model dominates across all crops. Standard LSTM achieves the highest average forecast accuracy (92.93%) and performs best for cassava and corn. Transformer excels for green bean (93.50%) and soybean (94.18%). ARIMA performs notably poorly for cassava (77.74%) due to structural breaks violating stationarity assumptions. Rolling-Window LSTM, with window sizes selected via validation rather than test-set feedback, achieves a lower average accuracy (82.85%) than the other models. This demonstrates the effect of proper hyperparameter selection: the original window sizes, which were tuned using test-year feedback, produced artificially inflated accuracy figures.

1) *Peak Timing Accuracy*: While Rolling-Window LSTM has a higher average MAPE than Standard LSTM with proper validation-based window selection, its key advantage is peak timing accuracy (Table VII).

¹ Defined as the complement of MAPE, not to be confused with classification accuracy.

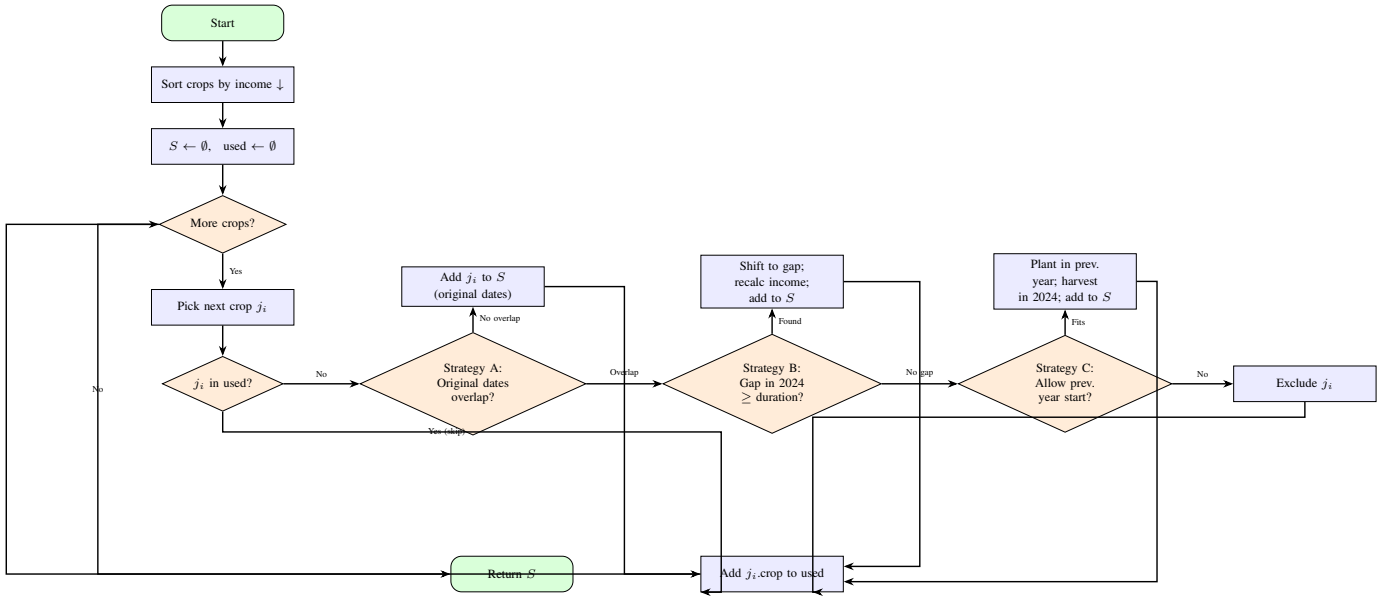


Fig. 8. Flowchart of the Greedy Crop Scheduling algorithm. Crops are processed in descending income order. For each crop, three placement strategies are attempted sequentially: (A) original peak-optimal dates, (B) shifted to an available gap within the target year, (C) planting in the previous year with harvest in the target year. If none succeeds, the crop is excluded.

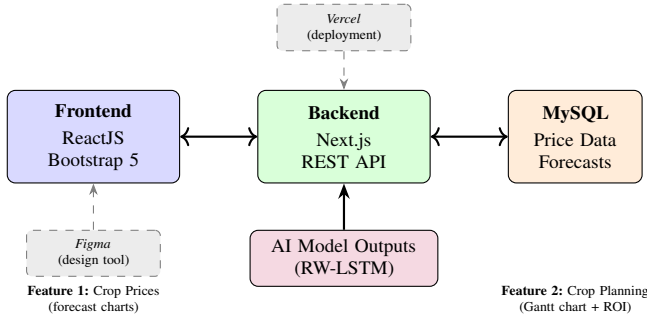


Fig. 9. Web application architecture. The frontend communicates with a Next.js backend connected to MySQL for data storage. AI model outputs feed into the backend for real-time recommendations.

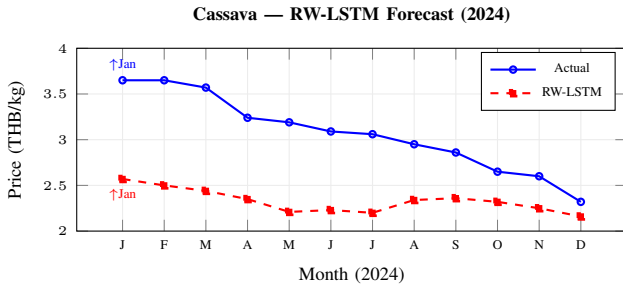


Fig. 10. RW-LSTM forecast for cassava (best peak timing). Peak correctly predicted in Jan (actual: Jan, predicted: Jan) with $\Delta\text{Month} = 0$.

Corn — RW-LSTM Forecast (2024)

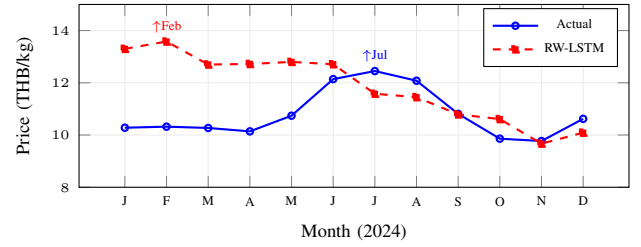


Fig. 11. RW-LSTM forecast for corn (challenging case). The predicted peak (Feb) arrives five months before the actual peak (Jul), with $\Delta\text{Month} = 5$.

TABLE VII
PEAK TIMING ACCURACY — ROLLING-WINDOW LSTM

Crop	Actual	Pred.	$\Delta\text{Mo.}$	ΔPrice	Rating
Cassava	Jan	Jan	0	29.57%	Excellent
Corn	Jul	Feb	5	9.12%	Poor
Green Bean	Sep	Sep	0	10.24%	Excellent
Soybean	Apr	Mar	1	20.44%	Excellent

Two of four crops (cassava and green bean) have exact peak month predictions, and soybean is within one month. Corn's peak prediction is five months off, reflecting the difficulty of capturing its mid-year price spike with a short (6-year) window. Note that the ΔPrice column reflects overall forecast error at the peak month, not scheduling quality: the scheduling system uses only the peak month position to set the harvest date, while income is computed from the predicted price at that month. Cassava's high ΔPrice (29.57%) despite exact peak timing indicates the forecast underestimates the price

level but correctly identifies *when* to harvest. Despite lower overall MAPE compared to the test-tuned configuration, the validated model retains strong peak timing for three of four crops—critical for the scheduling system.

B. Crop Planning Results

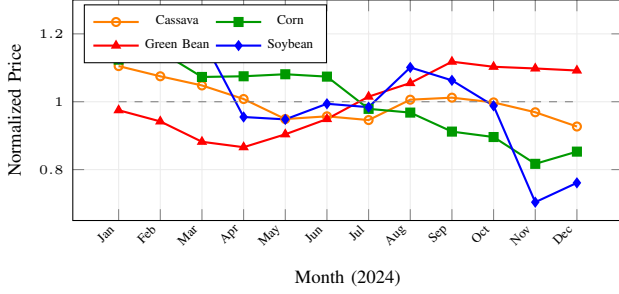


Fig. 12. Normalized RW-LSTM forecasts (2024). Each crop’s predicted prices are divided by their mean, enabling cross-crop comparison. Peaks occur in different months—Cassava (Jan), Corn (Feb), Soybean (Mar), Green Bean (Sep)—which enables non-overlapping crop rotation.

The price overlap analysis reveals that peak prices for the four crops occur in different months, enabling crop rotation. The unoptimized Gantt chart (Fig. 13), where each crop is placed at its peak-optimal position, shows that cassava, corn, and soybean overlap during October 2023–March 2024 due to their long growing periods extending into the previous year.

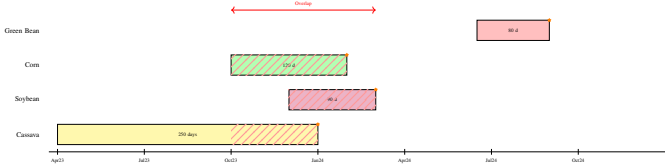


Fig. 13. Unoptimized Gantt chart. Each crop is placed at its peak-optimal planting–harvest window. Cassava, corn, and soybean overlap during Oct 2023–Mar 2024, indicated by hatched regions.

After applying exhaustive enumeration with the optimal ordering (Cassava → Soybean → Green Bean → Corn), all four crops were scheduled without overlap (Fig. 14). Corn and soybean were shifted to available gaps using Strategy B.

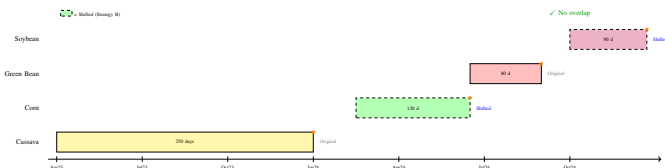


Fig. 14. Optimized Gantt chart (brute-force optimal ordering). Corn and soybean were shifted to available gaps using Strategy B (dashed borders), producing a non-overlapping four-crop rotation with total income of 39,212 THB/rai/year.

Table VIII details the income before and after optimization.

Corn and soybean lost income from shifting to non-peak months, while the total feasible income reached

TABLE VIII
INCOME COMPARISON: BEFORE VS. AFTER OPTIMIZATION

Crop	Before	After	Strategy	Δ
Cassava	8,998	8,998	Original	± 0
Corn	20,377	18,385	Shifted	−1,992
Green Bean	6,818	6,818	Original	± 0
Soybean	7,830	5,011	Shifted	−2,819
Total	<i>Infeasible</i>	39,212	—	—

39,212 THB/rai/year. This schedule was identified through exhaustive enumeration (Section V-B1) rather than the greedy heuristic. Table IX compares this with monoculture baselines.

TABLE IX
OPTIMIZED MULTI-CROP ROTATION VS. MONOCULTURE

Plan	THB/rai/yr	vs. Opt.
4-Crop Rotation	39,212	—
Mono: Corn (best)	20,377	+92.4%
Mono: Cassava	8,998	+335.8%
Mono: Soybean	7,830	+400.9%
Mono: Green Bean	6,818	+475.2%

The optimized four-crop rotation yields approximately 1.9 times the income of the best single-crop monoculture option (corn).

1) *Brute-Force Optimality Verification*: Exhaustive enumeration of all $4! = 24$ crop permutations was performed to verify the greedy heuristic. The brute-force search identified an optimal schedule yielding 39,212 THB/rai/year, compared to the greedy result of 33,394 THB/rai/year—an optimality gap of 14.8%. The greedy heuristic, by prioritizing corn (highest income) first, blocked soybean from being scheduled. In contrast, the optimal ordering (Cassava → Soybean → Green Bean → Corn) accommodates all four crops through strategic shifting. This result demonstrates that greedy scheduling is suboptimal even for four crops and motivates the use of exact methods such as exhaustive enumeration or Dynamic Programming for production deployment. Table VIII and Fig. 14 present the brute-force optimal schedule.

2) *Sensitivity to Yield Variability*: The income model assumes deterministic yields, but actual farm output varies due to weather, pest pressure, and management quality. To quantify the robustness of the rotation advantage, we conducted a sensitivity analysis by uniformly scaling all crop yields (Table X).

TABLE X
SENSITIVITY OF ROTATION INCOME TO YIELD VARIABILITY

Yield Factor	4-Crop (THB/rai/yr)	vs. Best Mono
1.1 (+10%)	43,133	$\times 1.92$
1.0 (baseline)	39,212	$\times 1.92$
0.9 (−10%)	35,291	$\times 1.92$
0.8 (−20%)	31,370	$\times 1.92$
0.7 (−30%)	27,448	$\times 1.92$

Income = Yield $\times$ Price, uniform scaling preserves the rotation-to-monoculture ratio ($\times 1.92$) across all yield factors.

Even at a 30% yield reduction—representative of a severe drought year—the four-crop rotation retains a significant income advantage over monoculture, demonstrating the structural benefit of temporal diversification. Because both rotation and monoculture incomes scale proportionally with yield, the relative advantage ($\times 1.92$) is invariant to uniform yield shocks. Differential yield shocks—where one crop fails but others survive—would alter this ratio and represent a scenario where rotation provides risk diversification beyond income maximization. However, these figures remain upper-bound estimates as production costs, post-harvest losses, and price-yield correlations are not modeled.

C. Web Application

The CropPlanner web application provides two primary interfaces for farmers, as shown in Figs. 15 and 16.

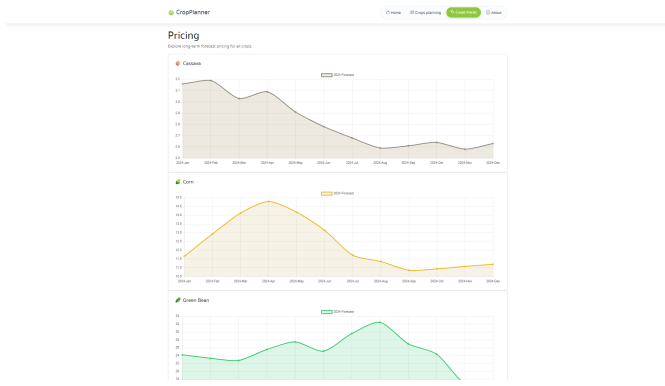


Fig. 15. CropPlanner — Crop Prices Page. Predicted price charts for all four crops with monthly granularity.

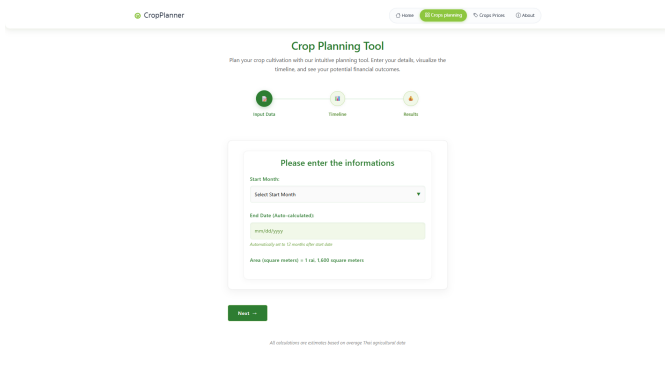


Fig. 16. CropPlanner — Crop Planning Page. Farmers input start month and area; the system returns a Gantt chart with financial summary (income, expenses, profit, and ROI).

The **Crop Prices** page displays Rolling-Window LSTM forecast charts for all four crops, enabling farmers to compare predicted price trends across a 12-month horizon. The **Crop Planning** page accepts a start month and plot area as inputs, runs the scheduling algorithm, and returns a Gantt chart with an accompanying financial summary—including projected income, estimated expenses, net profit, and ROI. The current prototype

uses the greedy heuristic; incorporating the exhaustive search—trivial for four crops (<1 ms)—is planned for the next release.

The application was deployed on Vercel with the Next.js backend serving REST API endpoints that query pre-computed RW-LSTM forecasts stored in MySQL. The interface is responsive across desktop and mobile devices, ensuring accessibility for farmers in rural areas with limited computing resources.

VI. DISCUSSION

A. No Single Model Dominates

No single model is best for all crops, attributable to the limited dataset (~ 252 points) and differing price dynamics. LSTM avoids overfitting and leads for cassava and corn; Transformer captures complex seasonal patterns for soybean; ARIMA fails for cassava (24.71%) due to stationarity violations but remains competitive elsewhere.

B. Peak Timing Over Overall MAPE

With validation-based window selection, RW-LSTM’s average MAPE (17.06%) is notably higher than Standard LSTM (7.07%), indicating that its original performance was inflated by test-set feedback—underscoring the importance of rigorous train-validation-test separation. However, its peak timing remains strong for three of four crops (cassava and green bean exact, soybean ≤ 1 month off), making it viable for scheduling where peak position matters more than pointwise accuracy. Corn’s 5-month error suggests its 6-year validated window is too short. An ensemble selecting the best model per crop would be ideal.

C. Greedy Scheduling: Verification and Scalability

With only four crops, the scheduling problem admits $4! = 24$ permutations, making exhaustive search trivial. Our brute-force verification revealed that the greedy heuristic is suboptimal: it produced 33,394 THB/rai/year (3 crops, soybean excluded), compared to the optimal 39,212 THB/rai/year (all 4 crops)—a gap of 14.8%. The greedy algorithm’s income-first prioritization placed corn first, which consumed a time slot that prevented soybean from being scheduled. The optimal ordering prioritizes cassava first (long growing period, must start early), followed by crops that fill the remaining gaps. This demonstrates that even with four crops, the scheduling structure is non-trivial: greedy approaches can miss solutions that require counter-intuitive priority orderings. For larger crop sets (>10), exhaustive search becomes infeasible ($15! \approx 1.3 \times 10^{12}$), and Dynamic Programming in $O(n \log n)$ becomes necessary.

D. Experimental Design and Income Model

A single test year (2024) limits generalizability; a dedicated validation year (2023) was used for RW-LSTM window size selection to prevent data leakage, and network hyperparameters were fixed *a priori*. The income model (Eq. 5) uses fixed yield averages and omits production costs and price-yield correlations, producing upper-bound gross revenue estimates. The sensitivity analysis (Section V-B2) confirms the rotation advantage is robust to uniform yield reductions, but a full economic model would provide more actionable guidance.

VII. LIMITATIONS AND FUTURE WORK

This study has several limitations: (1) a single test year (2024) limits the generalizability of model comparisons, though a validation year (2023) was used for hyperparameter selection to prevent data leakage; (2) no field testing with actual farmers was conducted, leaving usability and adoption unverified; (3) the income model uses fixed yield averages and omits production costs, weather-induced yield variability, and price-yield correlations, making reported income figures upper-bound estimates of gross revenue; (4) external factors such as government policy changes and global commodity prices are not incorporated into the forecasting models; and (5) coverage is limited to Northeastern Thailand and four field crops.

Future directions include: (1) ensemble model selection per crop to leverage each model's strengths; (2) multi-year rolling-origin evaluation (e.g., testing across 2022–2024) to provide stronger evidence of model stability; (3) incorporating weather, macroeconomic, and supply-chain features as additional input variables; (4) a full economic model with stochastic yields, production costs, and price-yield correlations for more realistic income estimates; (5) Dynamic Programming for scheduling with >10 crops; (6) user studies and pilot deployments with farmers to evaluate real-world impact; and (7) expanding to additional regions and crop categories including horticultural crops.

VIII. CONCLUSION

This paper presented CropPlanner, an integrated system for AI-driven crop planning that bridges the gap between price forecasting research and actionable agricultural recommendations.

First, a comparative study of four forecasting models revealed that no single model dominates. Standard LSTM achieved the highest average accuracy (92.93%), while Rolling-Window LSTM—with window sizes validated on a held-out year to prevent data leakage—retained accurate peak timing for three of four crops, underscoring the importance of proper experimental methodology and application-aligned metrics.

Second, exhaustive enumeration identified the optimal non-overlapping four-crop rotation yielding 39,212 THB/rai/year—approximately 1.9 times the best monoculture income. The greedy heuristic was suboptimal by 14.8%, demonstrating the value of exact methods. Sensitivity analysis confirmed robustness to uniform yield variability.

Third, the CropPlanner web application enables farmers to plan up to 12 months ahead through intuitive visualizations, delivering AI-powered insights without requiring technical expertise. This system has the potential to reduce income volatility and promote sustainable agriculture for Thai farmers.

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